

The Cost of Caution: Optimal Monetary Policy and the Value of Active Learning

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Abstract

This paper challenges the “Attenuation Principle” (Brainard, 1967) in the context of structural model uncertainty. While conventional wisdom suggests policymakers should act cautiously when facing uncertainty, we demonstrate that a passive (“myopic”) approach significantly reduces the signal-to-noise ratio, prolonging model ambiguity. By solving the dual control problem using a high-fidelity Finite Difference Method on a 125^3 grid, we identify a distinct “probing wedge”—an optimal deviation where the policymaker induces strategic volatility to accelerate belief convergence. We show that this “anti-attenuation” strategy generates a welfare gain reaching up to a 0.73 percentage point reduction in inflation volatility during periods of peak uncertainty, proving that the long-run value of model identification outweighs the short-run costs of stabilization.

JEL Classification: E52, E58, D83

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1 Introduction

Differing from the traditional “Attenuation Principle” of Brainard (1967) (which recommends a “cautious” response under exogenous multiplicative uncertainty), this paper demonstrates that an “aggressive” policy that strategically induces volatility is optimal under structural ambiguity when belief process is endogenous and the ambiguity can be resolved over time through learning. To rationalize this, note that since the central bank can affect economic outcomes, it can actively experiment the economy, gather data, and conduct Bayesian learning to resolve the structural ambiguity and find the true model. we show that the incentive

to resolve the ambiguity faster encourages the central bank to be more aggressive instead of more cautious, as being more aggressive can help the bank obtain clearer and more informative signals about the economy to help with Bayesian update.

Under Brainard (1967)’s classic multiplicative uncertainty, a “cautious” policy is mathematically defined as a proportional shrinkage of the policy instrument relative to the certainty-equivalent baseline, a necessary trade-off to minimize endogenous outcome variance. In contrast, in our setup, when uncertainty is structural and resolvable via Bayesian filtering, we define an “aggressive” monetary policy as the deliberate amplification of the instrument magnitude beyond the optimal level needed to stabilize inflation and output gap. As per Wieland (2000)’s policy experimentation concept, we offer a specific numerical solution by using a high fidelity Finite Difference Method (FDM) to the continuous time Hamilton-Jacobi-Bellman equation. Our main contribution is to solve the complete dual control problem (stabilizing the economy while controlling the speed of the resolution of ambiguity), quantify the optimal policy, and reveal significant welfare advantages of a proactive “probing” of the economy’s true structure.

For setup of the model in this paper, we focus on the structural ambiguity of the Price Phillips Curve, because there is a major ongoing debate about how people actually form inflation expectations (Galí and Gertler, 1999; Mankiw and Reis, 2002; Coibion and Gorodnichenko, 2015). So, we set up a continuous-time model where the central bank doesn’t know the true data-generating process of the economy by the true Philip Curve. It has to figure out if the economy is driven by a Backward-Looking (BL), adaptive Phillips curve, or a Forward-Looking New Keynesian one (NK). The policy instrument available to the central bank is the nominal interest rate, and the bank receives continuous, noisy signals from the economy, like inflation and the output gap. After receiving the signals, the Central bank use Continuous Bayesian Filtering “Kushner Equation” (derived in Lemma A.1, Appendix A) to update its belief state over time. This way we can analytically express the belief state

as a stochastic differential equation (SDE) $dp_t = \underbrace{p_t(1-p_t)\mathcal{H}(\mathbf{K}_t)}_{\sigma_p(\mathbf{K}_t, p_t)} d\bar{Z}_t^\pi$. (see Equation (6))

It is a martingale with no drift but with a diffusion term, that is positively related to the signal quality of inflation data. This means that if the signal quality of inflation data is higher, central banks' belief diffuse faster, thus it can speed up the learning and resolve the ambiguity faster.

The central mechanism behind how a more aggressive central bank can speed up the resolution of the ambiguity is characterized by the “probing wedge.” We define the probing wedge as the optimal deviation from a passive, “myopic,” benchmark. Under a myopic paradigm, the central bank only set the optimal interest rate needed to stabilize the economy. It acts stylistically in accordance with the original Brainard framework: it considers the structural belief state and its associated volatility as an exogenous stochastic process, and believe that their policy actions do not affect the rate at which uncertainty is resolved. On the contrary, an active policymaker recognizes that the quality of macroeconomic data is endogenous to their policy stance. Therefore, the policymaker uses the probing wedge to strategically steer the output gap into areas where the competing models — in this case the Backward-Looking (BL) and New Keynesian (NK) Price Philip Curves — give maximal conflicting predictions about inflation. The magnitude of the wedge is substantial and represents policy deviations ranging from 25 to 200 basis points relative to standard inflation bands, and theoretically can exceed 1000 basis points at the extreme boundaries of the state space (see Figure 1).

The basic motivation to probe derives from the cost of structural ambiguity itself. Since the BL and NK models require fundamentally different optimal interest rates, a policymaker who applies a single “compromise” policy during a period of structural ambiguity creates an excess expected stabilization cost. Thus, the policymaker's value function is strictly concave in beliefs, and being trapped in a state of high structural ambiguity is strictly more expensive than having knowledge of the true model. In order to escape this costly uncertainty, the central bank introduces a purposeful causal chain to deliberately produce information.

By departing from the myopic benchmark, the probing wedge strategically stretches the output gap. Since the competing models are in greatest disagreement regarding how inflation responds to the output gap, the targeted distortion produces the maximum signal intensity of the incoming inflation data $\mathcal{H}(\mathbf{K}_t)$. The stronger signal, in turn, increases the volatility of the central bank’s own structural beliefs (σ_p). This relationship is mathematically captured by the active learning term $\frac{1}{2}\sigma_p^2\frac{\partial^2 V}{\partial p^2}$ in the central bank’s HJB equation (see Equation (10)). Since the value function has negatively curvature, maximizing this belief variance creates a positive welfare effect. Instead of fearing volatility characterized by Brainard (1967), the active policymaker takes advantage of it: by deliberately maximizing belief volatility, the central bank accelerates the stochastic evolution of its posterior probability, allowing the economy to move away from the costly area of maximum uncertainty and toward the cost minimizing boundaries of full model identification.

Quantitatively, the welfare benefits derived from implementing the active learning strategy are large. The optimal policy provides a welfare benefit equal to an average 0.21, max 0.73 percentage point decrease in inflation volatility compared to a passive myopic policy (see Figure 2). Of course, the active learning policy is not a “free lunch”; it intentionally incurs a short term stabilization cost to purchase long-term clarity. This intertemporal tradeoff — sacrificing minimal stability in the present to prevent major policy mistakes in the future — is the core of the paper’s recommendation. The long run welfare gains from proactive experimenting with the economy are significantly larger than the short run costs.

As we have explored above, the aggressive probing is not simply for any type of information, and the analysis in our paper shows that it is a targeted, risk-driven decision-making process. The degree of probing is not constant and raises questions: What specific risk causes the policymaker to act so aggressively? Our analysis of the sensitivity of the policy to changes in the characteristics of the model reveals a surprising factor that drives the level of aggressiveness of the policy: the incentive to explore increases exponentially when the NK Phillips Curve slope is close to zero ($\kappa_{NK} \rightarrow 0$; see Figure 6), i.e., when one of the potential

models suggests that the policymaker’s tools are almost powerless. This shows a strong precautionary motivation to avoid policy impotence. The central bank fears being “trapped” in a world in which its tools are completely ineffective, and its active probing is motivated by the *hope* of finding a controllable economy. This is in direct contrast to the “ambiguity aversion” of Hansen and Sargent (2008), in which the aggressive policy is motivated by the *fear* of a worst-case misspecification.

The challenge of optimal policy under structural uncertainty has typically been framed as a conflict between caution and aggressiveness. The original “Attenuation Principle” Brainard (1967) suggests that policymakers should be less aggressive when dealing with exogenous multiplicative uncertainty in order to limit the increase in the variance of outcomes. Conversely, the Active Learning literature Wieland (2000); Bertocchi and Spagat (1993); Ellison and Vilmunen (2005) suggests that a forward looking agent should proactively experiment in order to reduce posterior uncertainty and accept the short term output volatility required to achieve long term model identification. A related strand, the Robust Control framework Hansen and Sargent (2008), advocates for an aggressive policy not to learn, but to insure against worst-case misspecifications. Using continuous time methods Achdou et al. (2022); Nuño and Thomas (2022) we bridge these approaches. We show that the policymaker experiments not due to ambiguity aversion, but to efficiently collapse the probability distribution onto the true model. Although continuous Bayesian updating guarantees that the true model is eventually revealed, thereby avoiding the absolute “information trap” of permanent ignorance seen in examples like Berentsen et al. (2008) — a passive policy severely degrades the signal-to-noise ratio, trapping the economy in a long period of structural ambiguity that depletes welfare.

2 The Economic Environment

We develop a continuous-time model of monetary policy under structural uncertainty, where a central bank faces a trade-off between short-term stabilization and long-term learning. The policymaker does not know whether inflation dynamics are governed by a Backward-Looking (BL) or a New Keynesian (NK) process, but acts as a Bayesian learner, updating beliefs by filtering the continuous stream of economic data. We assume the true model $m \in \{0, 1\}$ is fixed but unknown, where $m = 0$ denotes the BL model and $m = 1$ denotes the NK model.

2.1 Structural Models and Filtration

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying a two-dimensional standard Brownian motion $\mathbf{Z}_t = (Z_t^\pi, Z_t^x)^\top$. The physical state of the economy is denoted by the vector $\mathbf{K}_t = (\pi_t, x_t)^\top$. The policymaker observes the filtration generated by the state history, $\mathcal{F}_t = \sigma(\mathbf{K}_s : s \leq t)$.

Under each hypothesis, the economy evolves according to a specific stochastic differential equation (SDE) driven by these independent shocks:

Model 0 (Backward-Looking): We model backward-looking (BL) adaptive expectations as a mean-reverting continuous-time process driven by the output gap. While stylized compared to a discrete-time DSGE framework, this continuous-time formulation cleanly isolates the “sticky” inertia characteristic of traditional adaptive models:

$$d\pi_t = \lambda_{BL}(\kappa_{BL}x_t - \pi_t)dt + \sigma_\pi dZ_t^\pi. \quad (1)$$

Model 1 (New Keynesian): In contrast to a fully micro-founded Rational Expectations Equilibrium (which strictly requires agents to already know the true structural parameters), we define the New Keynesian (NK) regime as a *Perceived Law of Motion (PLM)*. This stylized continuous-time formulation captures the canonical forward-looking dynamics—specifically, the fundamental instability induced by the positive discount factor $\rho_{NK} > 0$ on current inflation. By contrasting this explosive tendency with the mean-reverting

BL model, we capture the core identification challenge: distinguishing between an inherently self-stabilizing economy and one requiring active anchoring. The drift follows:

$$d\pi_t = (\rho_{NK}\pi_t + \kappa_{NK}x_t) dt + \sigma_\pi dZ_t^\pi. \quad (2)$$

where $\rho_{NK} > 0$ represents the destabilizing effect of unanchored expectations.

The output gap dynamics are model-independent and follow a standard IS curve specification:

$$dx_t = \lambda_x(-x_t - \gamma(i_t - \pi_t)) dt + \sigma_x dZ_t^x. \quad (3)$$

2.2 Information Structure and Belief Dynamics

The central bank's belief that the true model is the NK model is denoted by the posterior probability $p_t \equiv p_{NK,t} = \mathbb{P}(m = 1|\mathcal{F}_t)$. The evolution of this belief is governed by the filtering of the inflation signal.

We define the *Signal Intensity* function $\mathcal{H} : \mathbb{R}^2 \rightarrow \mathbb{R}$ as the standardized distance between the drift components of the two competing models:

$$\mathcal{H}(\mathbf{K}_t) \equiv \frac{\mu_\pi^{\text{NK}}(\mathbf{K}_t) - \mu_\pi^{\text{BL}}(\mathbf{K}_t)}{\sigma_\pi}. \quad (4)$$

This term represents the instantaneous signal-to-noise ratio available to the policymaker.

To characterize the belief dynamics, we introduce the *inflation innovation process* $d\bar{Z}_t^\pi$, which represents the “surprise” component of the observed inflation under the policymaker's filtration:

$$d\bar{Z}_t^\pi \equiv \frac{1}{\sigma_\pi} (d\pi_t - \bar{\mu}_\pi(p_t, \mathbf{K}_t)dt). \quad (5)$$

where $\bar{\mu}_\pi(p_t, \mathbf{K}_t) = p_t\mu_\pi^{\text{NK}} + (1-p_t)\mu_\pi^{\text{BL}}$ is the belief-weighted expected drift. By the Kushner-Stratonovich equation (derived in Lemma A.1, Appendix A), the belief process evolves as a

martingale driven by these innovations:

$$dp_t = \underbrace{p_t(1 - p_t)\mathcal{H}(\mathbf{K}_t)}_{\sigma_p(\mathbf{K}_t, p_t)} d\bar{Z}_t^\pi. \quad (6)$$

This stochastic differential equation is central to our framework, as it cleanly isolates the fundamental mechanics of the active learning channel. The endogenous belief volatility, $\sigma_p(\mathbf{K}_t, p_t) = p_t(1 - p_t)\mathcal{H}(\mathbf{K}_t)$, is governed by three distinct components that dictate the speed of structural identification:

First we have the baseline uncertainty bound $p_t(1 - p_t)$. This parabolic term represents the variance of a Bernoulli trial. Economically, this dictates that the speed of learning is naturally maximized when structural ambiguity is at its peak ($p_t = 0.5$). As beliefs converge toward the full-information boundaries (0 or 1), the Bayesian updating process smoothly decelerates, naturally preventing the posterior probability from breaching its logical limits.

Second we have the endogenous signal intensity $\mathcal{H}(\mathbf{K}_t)$. Defined as the standardized distance between the drift components of the two competing models, $\mathcal{H}(\mathbf{K}_t) \equiv (\mu_\pi^{\text{NK}}(\mathbf{K}_t) - \mu_\pi^{\text{BL}}(\mathbf{K}_t))/\sigma_\pi$, this term captures the instantaneous signal-to-noise ratio. This is the exact mathematical channel through which the central bank exerts control over the flow of information. Because the competing Price Philip Curve models' inflation drift predictions diverge most sharply based on the output gap, the policymaker can use the interest rate to deliberately stretch x_t , forcing the models into conflict and artificially maximizing the signal quality.

Third we have the structural innovation $d\bar{Z}_t^\pi$. The diffusion is driven purely by the innovation term, confirming that structural beliefs only shift when incoming inflation data fundamentally deviates from the weighted theoretical expectation.

By decomposing the filtering equation analytically in this manner, the mathematical elegance of the continuous dual control problem becomes apparent. The central bank cannot predict the direction of the next belief update, as dp_t is a martingale, but it has direct leverage

over the magnitude of the update's variance via the signal intensity $\mathcal{H}(\mathbf{K}_t)$. Maximizing this endogenous variance to accelerate the resolution of ambiguity is the core objective of the probing wedge of the nominal interest rate.

2.3 The Planner's Problem

The central bank seeks to minimize the expected discounted quadratic loss over an infinite horizon. We define the augmented state vector as $\mathbf{X} = (\pi, x, p)^\top \in \mathbb{R}^2 \times [0, 1]$. The value function $V(\mathbf{X})$ is defined as:

$$V(\mathbf{X}) = \min_{\{i_t\}} \mathbb{E} \left[\int_0^\infty e^{-\rho t} \mathcal{L}(\pi_t, x_t, i_t) dt \right]. \quad (7)$$

where the instantaneous loss function $\mathcal{L}$ is:

$$\mathcal{L}(\pi, x, i) = \pi^2 + \omega_x x^2 + \omega_i (i - i_{\text{target}})^2. \quad (8)$$

2.4 The Hamilton-Jacobi-Bellman Equation

The value function $V(\mathbf{X})$ satisfies the Hamilton-Jacobi-Bellman (HJB) equation:

$$\rho V(\mathbf{X}) = \min_i \{ \mathcal{L}(\pi, x, i) + \mathcal{A}[i]V(\mathbf{X}) \}. \quad (9)$$

where $\mathcal{A}[i]$ is the infinitesimal generator for the augmented state process. By applying Itô's formula to the system coupled with the belief martingale (see Appendix A for the full derivation), we obtain the decomposed dual control equation:

$$\begin{aligned} \rho V(\mathbf{X}) = \min_i \left\{ \mathcal{L} + \mathcal{A}^{\text{myopic}}[i]V + \mathcal{T}_{\text{belief}} \right\} \\ \text{where } \mathcal{T}_{\text{belief}} \equiv \underbrace{\frac{1}{2}\sigma_p^2 V_{pp}}_{\text{Value of Information}} + \underbrace{\sigma_p \sigma_\pi \rho_{\pi p} V_{\pi p}}_{\text{Learning-Shock Correlation}} \end{aligned} \quad (10)$$

where $\mathcal{A}^{\text{myopic}}[i]$ represents the standard infinitesimal generator for the physical state dynamics, treating the belief state as an exogenous process. The term involving V_{pp} (typically negative) captures the welfare gain from actively resolving uncertainty, providing the rigorous foundation for the “probing wedge.”

2.5 The Value of Active Learning and the Probing Wedge

To isolate the experimentation incentive, we decompose the value function into a myopic component V^M and the *Value of Active Learning* $\mathcal{I}$:

$$V(\mathbf{X}) = V^M(\mathbf{X}) - \mathcal{I}(\mathbf{X}). \quad (11)$$

The myopic solution V^M is formally defined as the solution to a restricted version of the HJB equation (10) where the active learning channel is shut down ($\mathcal{T}_{\text{belief}} = 0$):

$$\rho V^M(\mathbf{X}) = \min_i \left\{ \mathcal{L}(\pi, x, i) + \mathcal{A}^{\text{myopic}}[i] V^M(\mathbf{X}) \right\}. \quad (12)$$

The myopic solution V^M corresponds to the classic benchmark established by Brainard (1967). In the standard Brainard framework, the variance of the policy multiplier is assumed to be strictly exogenous and unalterable. Faced with this fixed multiplicative uncertainty, a static planner minimizing a quadratic loss function has only one mathematical recourse to reduce outcome variance: attenuate the magnitude of the policy instrument. Our myopic generator ($\mathcal{A}^{\text{myopic}}$) perfectly replicates this logic within a continuous-time setting. By nullifying the active learning operator ($\mathcal{T}_{\text{belief}} = 0$), the myopic agent treats the structural belief state p_t and its associated volatility σ_p as an exogenous stochastic process. Because it cannot strategically manipulate the signal-to-noise ratio to resolve future ambiguity, the myopic agent simply minimizes instantaneous expected loss subject to the current parameter uncertainty, inevitably defaulting to the conservative, attenuated policy stance.

Proposition 1 (The Probing Wedge). The optimal active monetary policy i_{active}^* can

be decomposed into a myopic stabilization term and a probing wedge:

$$i_{\text{active}}^*(\pi, x, p) = \underbrace{i_{\text{myopic}}^*(\pi, x, p)}_{\text{Stabilization}} + \underbrace{\Delta_{\text{probe}}(\pi, x, p)}_{\text{Experimentation}}. \quad (13)$$

where the probing wedge is given by (see Appendix A for derivation):

$$\Delta_{\text{probe}} = -\frac{\lambda_x \gamma}{2\omega_i} \frac{\partial \mathcal{I}}{\partial x}. \quad (14)$$

The probing wedge Δ_{probe} represents the optimal deviation from the certainty-equivalent rule. Intuitively, this deviation is driven by the marginal value of information with respect to the output gap ($\frac{\partial \mathcal{I}}{\partial x}$). By adjusting the interest rate, the central bank intentionally manipulates the state x_t to maximize the flow of information (via σ_p) and capture the value of active learning.

2.6 Theoretical Properties: Concavity and Anti-Attenuation

The fundamental driver of probing is the concavity of the value function in beliefs.

Proposition 2 (The Cost of Structural Uncertainty). *Let $V^{NK}(\pi, x)$ and $V^{BL}(\pi, x)$ be the boundary value functions. For any $p \in (0, 1)$, the myopic value function satisfies $V^M > V^{FI}$, where $V^{FI} = pV^{NK} + (1 - p)V^{BL}$ is the Full Information Benchmark. (See Appendix A for a formal proof).*

The economic intuition for this result is rooted in the suboptimality of a “compromise” policy. Because the optimal interest rate under the NK model differs from that under the BL model, applying a single averaged policy away from the steady-state generates an excess expected cost. This inherent cost of structural ambiguity makes the value function strictly concave in beliefs ($V_{pp} < 0$).

This concavity creates the “experimental motivation” identified by Wieland (2000). While Brainard (1967) suggests uncertainty should lead to caution, our framework shows that the

agent induces “strategic aggressive probing” to maximize signal intensity and escape the region of maximum uncertainty.

3 Numerical Methodology

We calibrate the model to standard values in the continuous-time monetary policy literature and solve the non-linear Hamilton–Jacobi–Bellman (HJB) equation using a high-fidelity Finite Difference Method (FDM) on a tensor grid of size $125 \times 125 \times 125$ ($N \approx 1.95 \times 10^6$ points). Given the lack of a closed-form solution, we employ a “False Transient” pseudo-time method with an upwinding scheme to ensure the discrete operator maintains the M-matrix property. This approach guarantees that our value function rigorously captures the “Information Dividend”—the curvature $\partial^2 V / \partial p^2 < 0$ —preventing non-physical oscillations and confirming that the optimal policy’s topology is an economic property rather than a numerical artifact. Detailed tables of baseline parameters and extensive discussion of the iterative solver (JAX/GMRES), operator formalization, and boundary conditions are provided in Appendix C.

4 Results

4.1 The Probing Wedge and the Geometry of Dual Control

The centerpiece of our numerical findings is the *Probing Wedge*, defined as the optimal deviation from the passive myopic Brainardian benchmark: $\Delta_{\text{probe}} \equiv i_{\text{active}}^* - i_{\text{myopic}}^*$. Figure 1 maps this wedge across the state space $[\pi, p]$ for $x \approx 0$. To ensure the veracity of these results, the underlying HJB equation is solved on a high-density 125^3 tensor grid. We utilize first-order upwinding for all drift terms to maintain the Monotonicity (M-matrix property) of the discrete operator, ensuring that the observed policy non-linearities are fundamental features of the dual control problem rather than numerical artifacts.

As derived in Proposition 1, the wedge is the central bank’s “demand for information,” driven by the gradient of the *Value of Active Learning* $\mathcal{I} = V^M - V$. Our solution reveals that the Value Function $V(\mathbf{K}, p)$ is strictly concave (hump-shaped) in belief space ($V_{pp} < 0$) due to the Cost of Structural Uncertainty (Proposition 2). In the context of the HJB, the term $\frac{1}{2}\sigma_p^2 V_{pp}$ in Equation (10) acts as a negative cost: by inducing an output gap that maximizes belief volatility σ_p , the agent leverages the concavity of the value function to push the state away from the expensive $p_{NK} = 0.5$ “hump” toward the cheaper, full-information boundaries ($p_{NK} = 0$ or 1).

This behavior constitutes strategic aggressive probing, or *Anti-Attenuation*. Figure 1 illustrates a stark anti-symmetry in the probing wedge across the inflation state space. When model ambiguity is high ($p_{NK} \approx 0.5$) and the economy faces positive inflation ($\pi_t > 0$), the central bank sets rates significantly higher than the myopic benchmark (the red region, reaching up to +1250 basis points). Conversely, in deflationary states ($\pi_t < 0$), the active policy prescribes aggressive rate cuts, dropping substantially below the certainty-equivalent baseline (the blue region, reaching −1250 basis points). This directional divergence reveals the mechanics of the active learning channel. The central bank does not blindly raise rates to learn; rather, it strictly *amplifies* the standard directional policy response. By pushing the interest rate to extremes, the agent induces a larger magnitude output gap x_t . Because the Signal Intensity $\mathcal{H}(\mathbf{K})$ relies on the distance between the forward-looking New Keynesian and the inertial Backward-Looking drifts, stretching the output gap artificially forces the two models to yield contrasting inflation predictions, thereby maximizing the signal-to-noise ratio. The intensity of this wedge explicitly tracks the topology of the value function. The peak magnitudes (deepest red and blue) concentrate around $p_{NK} \in [0.4, 0.6]$ and expand outward as the absolute magnitude of inflation increases. This is the exact region where the expected cost of structural uncertainty—and thus the marginal value of information driven by the concavity term $\frac{1}{2}\sigma_p^2 V_{pp}$ in Equation (10) is maximized.

It is important to note that the peak wedge of ≈ 1250 basis points occurs at the edges

of the state space (e.g., inflation deviations approaching $\pm 15\%$). To ensure these extreme policy swings do not reflect numerical artifacts from unconstrained control variables at the boundaries, we explicitly enforce structural policy constraints, clipping the nominal interest rate between -20% and $+50\%$. Even with these bounds securely in place, the probing wedge remains robust and substantial, demonstrating that the demand for information is a fundamental economic result. Our results demonstrate that the Brainard (1967) Attenuation Principle can be formally nested within the dual control framework as a corner solution. Specifically, Brainard-style attenuation dominates only when the informational signal-to-noise ratio $\mathcal{H}(\mathbf{K})$ approaches zero. In such limits, the endogenous belief volatility (σ_p) vanishes, neutralizing the value of active learning ($\mathcal{T}_{\text{belief}} \rightarrow 0$). The active Hamiltonian subsequently collapses to the myopic baseline, proving that standard attenuation is optimal only when the marginal cost of induced volatility strictly eclipses the marginal value of structural identification. This strategic deviation is not merely a tactical maneuver; it is the fundamental driver of the substantial welfare gains discussed in the next section, as the agent accelerates the transition of the economy toward the more efficient, low-cost boundary manifolds.

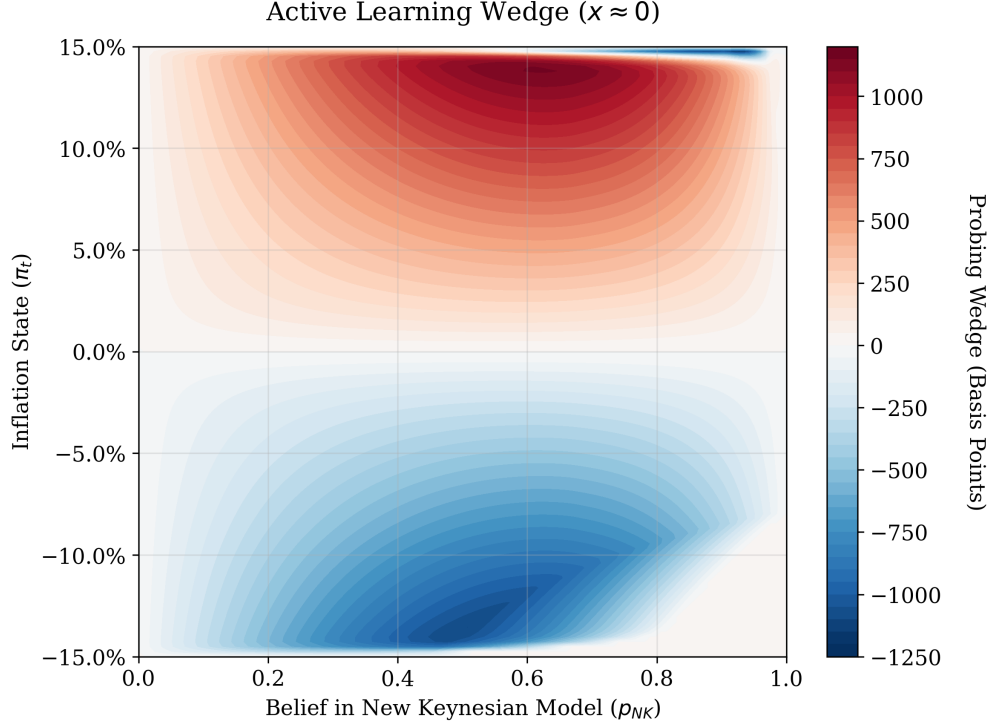


Figure 1: The Active Learning “Probing Wedge” ($\Delta_{\text{probe}} = i_{\text{active}}^* - i_{\text{myopic}}^*$) evaluated at output gap $x \approx 0$. The contour map demonstrates the optimal deviation from the myopic benchmark in basis points. Red regions ($\pi_t > 0$) indicate an active policy strictly tighter than the myopic benchmark, while blue regions ($\pi_t < 0$) indicate an aggressively looser, more accommodative stance. The magnitude of this wedge is substantial, representing policy deviations of 25-200 basis points within standard inflation bounds, and theoretically reaching over 1000 basis points at the extreme boundaries of the state space. This anti-symmetry shows the central bank amplifies its standard directional response at peak model ambiguity ($p_{NK} \approx 0.5$) to stretch the output gap and intentionally generate high-fidelity signals (Anti-Attenuation).

4.2 Welfare Gains: The Information Dividend

To quantify the value of active learning, we convert the value function improvement into a *Loss-Equivalent Volatility Reduction*, defined by the reduction in the root-mean-square discounted loss, scaled to percentage points: $\sigma_{\text{equiv}} = 100 \cdot (\sqrt{V^M} - \sqrt{V})$. This metric represents the effective reduction in economic volatility achieved by the active policy, assuming the loss is dominated by inflation variance. As shown in Figure 2, the welfare gain is substantial, reaching **up to 0.73 percentage points**. Interpretationally, this suggests that a failure

to learn effectively increases the “effective noise” of the environment significantly relative to the active baseline.

The unconditional expected welfare gain, evaluated via Monte Carlo integration over the transient learning dynamics (assuming an uninformative prior $p_0 = 0.5$ and steady-state initial physical conditions), is approximately 0.21 percentage points. This “Information Dividend” reflects the active policy’s ability to efficiently shift the economy’s probability mass away from regions of high structural ambiguity faster than the passive benchmark. As illustrated in Figure 2, the highest welfare gain (the brightest yellow band) is concentrated around the zero-inflation line ($\pi \approx 0$). This indicates that the value of active learning is maximized when the economy is relatively stable, providing a “safe” baseline from which to manage future shocks and efficiently learn. Conversely, the intensity smoothly tapers off to near zero (dark purple regions) as inflation approaches extreme high or low values ($\pm 15\%$). When inflation is far from the target, the immediate costs of stabilization dominate, leaving little room for discretionary experimentation.

It is crucial to contextualize this welfare metric. In standard heterogeneous-agent or representative-agent models (Lucas, 1987; Krusell and Smith, 1999), the pure consumer welfare gain from sub-percentage point reductions in inflation volatility is often calculated to be relatively small. However, our metric captures the minimization of the central bank’s quadratic loss function. For a monetary authority whose explicit institutional mandate is stabilization, an equivalent 20% relative reduction in baseline inflation volatility is massive. The primary welfare gain of the active policy is not merely the marginal smoothing of consumption, but the preservation of institutional credibility. By systematically escaping the “Information Trap,” the central bank prevents the massive, compounding accumulation of quadratic losses that inevitably occurs when policy is strictly optimized for the wrong structural model over a prolonged period.

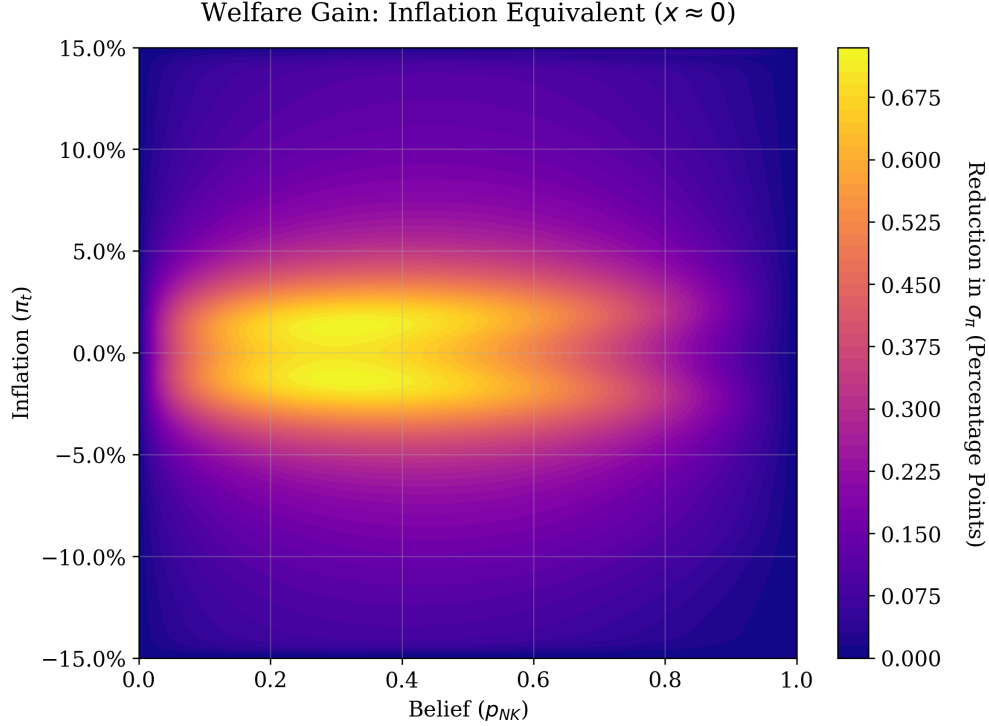


Figure 2: Welfare gain of the active learning policy, expressed as the equivalent reduction in inflation volatility (σ_π). The gain represents the difference between the myopic value function—which suffers from the excess expected cost of structural ambiguity ($V_{pp} < 0$)—and the active value function. The active policy strategically accelerates belief convergence toward the linear-quadratic manifolds at the full-information boundaries ($p_{NK} \in \{0, 1\}$). The bright yellow region indicates that the value of information is maximized near steady-state inflation ($\pi \approx 0$) at peak model ambiguity ($p_{NK} \approx 0.5$), whereas the dark purple regions show negligible gains where extreme inflation states make stabilization costs dominant.

4.3 Learning Dynamics and the Information Trap

To quantify the speed of structural identification, we simulate the learning dynamics of both the active and myopic controllers over a 12-year horizon. We run $N = 2000$ Monte Carlo paths, initializing the economy at the steady state ($\pi_0 = 0$, $x_0 = 0$) under maximum structural ambiguity ($p_0 = 0.5$). For this simulation, the New Keynesian specification is set as the true underlying data-generating process. Figure 3 tracks the evolution of the structural belief p_{NK} over time. The myopic baseline frequently falls into an “Information Trap.” By prioritizing immediate stabilization, the myopic policy aggressively neutralizes the

output gap ($x_t \approx 0$). Because the competing models' drift divergence is heavily dependent on the output gap, this optimal passive stabilization inadvertently suppresses the signal-to-noise ratio, stalling the filtration process. In contrast, the active policy strategically tolerates minor output gap deviations, maintaining a sufficiently high signal intensity $\mathcal{H}(\mathbf{K}_t)$ to ensure continuous belief updating.

To isolate the physical mechanics of the active learning channel, we compute the deterministic expected impulse response to a severe inflationary shock. The economy is initialized with an unanchored 5% inflation deviation ($\pi_0 = 0.05$) and zero output gap ($x_0 = 0$), again under maximum structural uncertainty ($p_0 = 0.5$). Figure 4 demonstrates how the central bank manipulates its policy instrument to purchase information. The active policymaker front-loads the stabilization effort, inducing a significantly sharper initial nominal rate hike than prescribed by the myopic rule. This aggressive initial policy stance deliberately stretches the output gap, maximizing the squared signal intensity $\mathcal{H}(\mathbf{K}_t)^2$ between the predicted inflation drifts of the two structural models. Critically, this demonstrates that the optimal active policy relies on *opportunistic probing* rather than the injection of random, unprompted volatility. A common critique of active learning is that central banks cannot politically justify artificially destabilizing a quiet economy to gather data. However, our framework reveals that the central bank does not need to manufacture shocks; instead, it leverages exogenous natural volatility. When an inflationary shock occurs, the active policymaker reacts asymmetrically, stretching the output gap dynamically during the stabilization process. The central bank trades a politically defensible, short-term excess stabilization effort for long-term structural clarity.

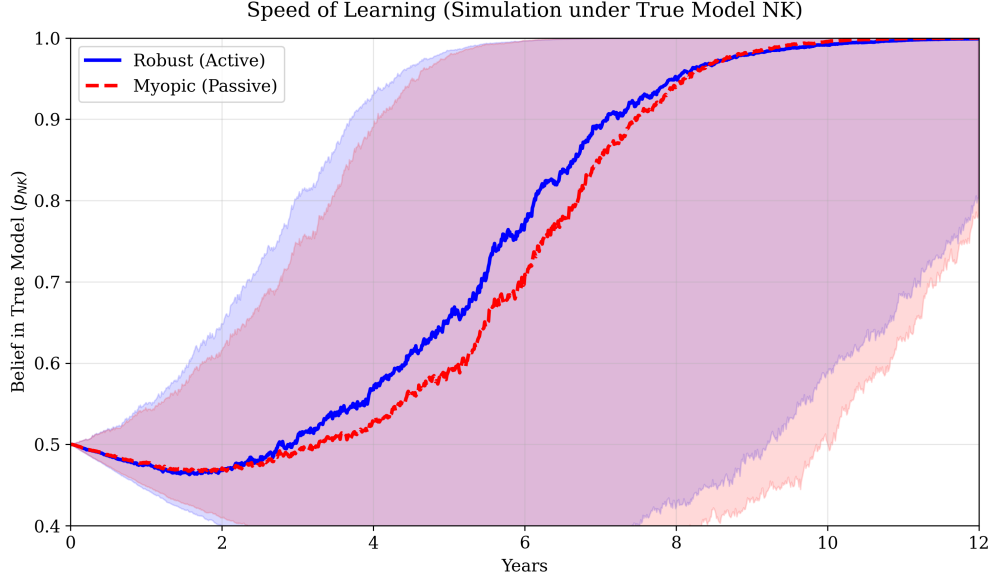


Figure 3: Monte Carlo Simulation of Belief Convergence. The figure plots the evolution of the policymaker’s belief p_{NK} over 12 years, assuming the New Keynesian model is the true data-generating process. The simulation consists of 2,000 paths initialized at the physical steady state ($\pi_0 = 0, x_0 = 0$) and maximum structural ambiguity ($p_0 = 0.5$). Solid and dashed lines represent the median belief trajectory for the robust (active) and myopic (passive) policies, respectively; shaded regions denote the interquartile range (25th to 75th percentiles). The myopic policy falls into an “Information Trap,” where passive stabilization drives the output gap near zero, collapsing the signal-to-noise ratio and stalling learning. The active policy strategically perturbs the economy to maintain high signal intensity, accelerating model identification.

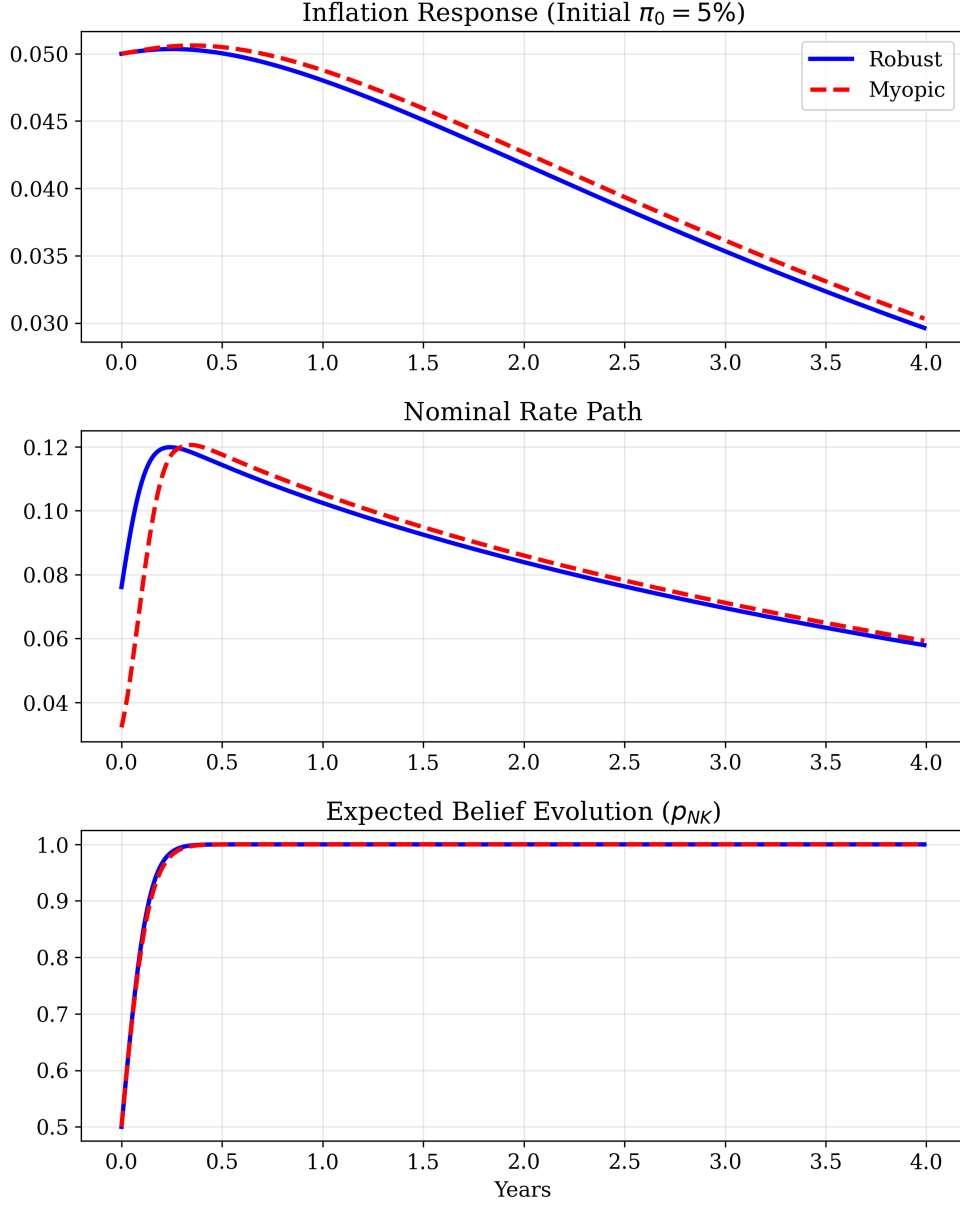


Figure 4: Expected Impulse Response to an Inflationary Shock. The figure illustrates the deterministic expected path of the economy following an initial 5% inflation shock ($\pi_0 = 0.05, x_0 = 0$) under maximum structural uncertainty ($p_0 = 0.5$). The true underlying model is New Keynesian. The middle panel highlights the active learning mechanism: the robust agent front-loads the policy response, raising the nominal interest rate higher and faster than the myopic benchmark. This initial divergence intentionally amplifies the output gap magnitude to maximize the signal-to-noise ratio in the filtered innovation process.

4.4 Performance Under a Structural Break

Figure 5 demonstrates the framework’s adaptability during an unannounced regime shift. We observe a latency period (from $t = 25$ to $t = 37$) where the cumulative signal must overcome the background noise σ_π . The speed of detection is governed by the *Signal Intensity* function $\mathcal{H}(\mathbf{K}) \equiv (\mu_\pi^{\text{NK}} - \mu_\pi^{\text{BL}})/\sigma_\pi$ (from Lemma A.1). Because the Active agent maintains “precautionary probing” even in low-volatility states, the signal intensity remains sufficiently high that once the shift is detected, beliefs converge with extreme rapidity. This “anti-attenuation” stance minimizes the recognition lag, preventing the massive accumulation of stabilization errors that plague the myopic controller post-break. This proactive learning acts as an insurance policy, accepting higher costs at $t < 25$ to ensure rapid detection and re-anchoring when the environment shifts at $t = 25$.

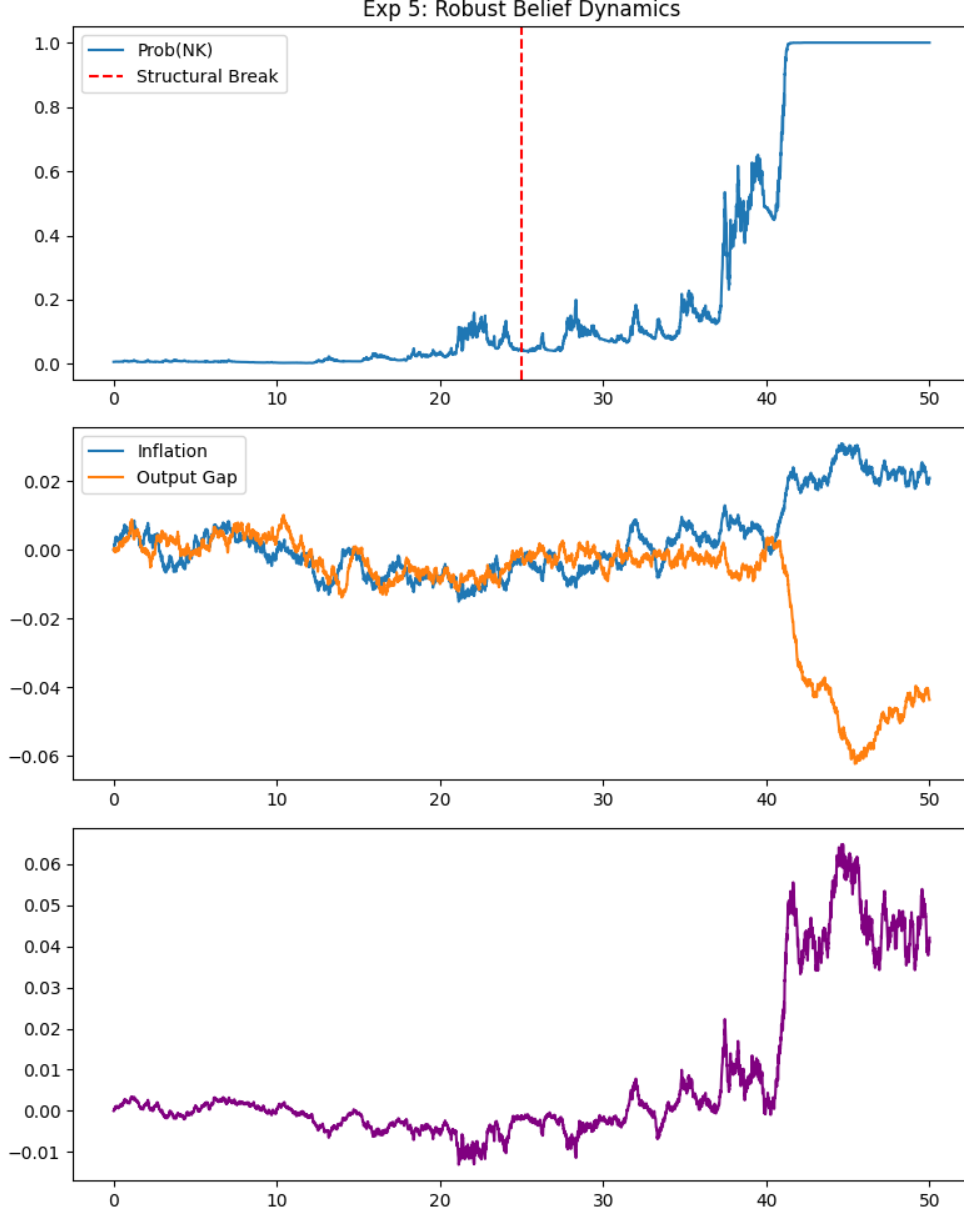


Figure 5: Dynamics of an Unannounced Structural Break. The figure illustrates a simulated regime shift from the Backward-Looking (BL) to the New Keynesian (NK) model at $t = 25$ (red dashed line). The top panel tracks the central bank's posterior belief of the NK model, p_t . The middle panel displays the physical state variables: inflation (π_t) and the output gap (x_t). The bottom panel plots the active policy instrument. Following the break, the active learning framework exhibits a necessary latency period ($t \in [25, 37]$) during which the cumulative signal intensity $\mathcal{H}(\mathbf{K}_t)$ must overcome exogenous background noise σ_π . By maintaining a continuous stance of precautionary probing, the active policy ensures rapid detection and belief convergence, re-anchoring the economy significantly faster than a passive myopic benchmark.

4.5 Comparative Statics: Drivers of the Wedge

To understand the mechanics governing the active learning policy, we conduct a sensitivity analysis over key structural parameters (detailed formally in Appendix B). We highlight three primary drivers:

First, *Signal-to-Noise Compensation*: Counter to the standard Attenuation Principle, as exogenous inflation noise (σ_π) increases, the active policymaker compensates by *increasing* the intensity of the probe. The value of information is so dominant that the agent is willing to accept higher stabilization costs to force a discernible signal through a noisy environment.

Second, *Precautionary Probing and the Asymmetry of Distance*: The wedge is not driven symmetrically by the structural “distance” between the models, but asymmetrically by the fear of policy impotence. The probing wedge becomes massive when the NK Phillips curve slope approaches zero ($\kappa_{NK} \rightarrow 0$). This confirms the central bank probes primarily to avoid being trapped in a “weak control” regime where its policy instrument is ineffective (see Figure 6).

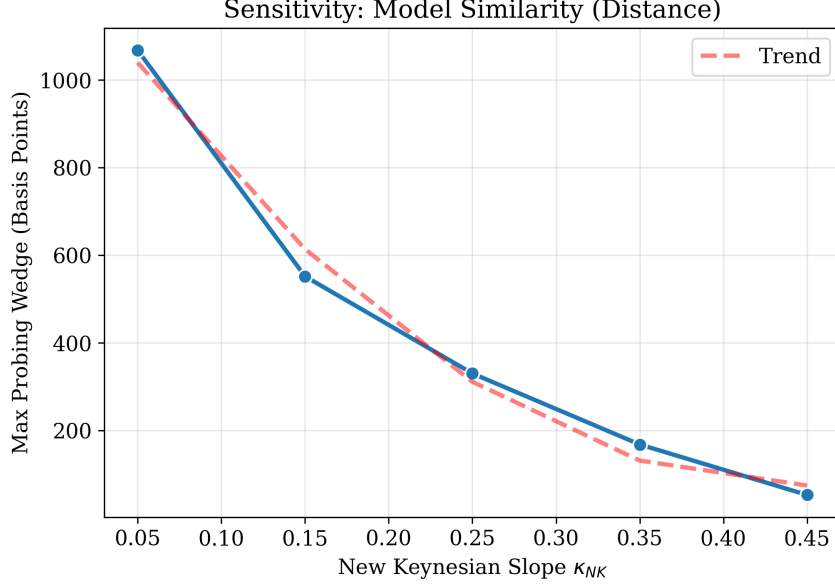


Figure 6: Sensitivity of the Probing Wedge to Model Similarity (κ_{NK}). The plot illustrates the maximum optimal deviation from the myopic benchmark (in basis points) as a function of the New Keynesian Phillips curve slope. Rather than scaling symmetrically with structural distance from Backward-Looking model, the probing wedge exhibits an explosive, non-linear acceleration as $\kappa_{NK} \rightarrow 0$. This singularity mathematically formalizes the transition into a “Weak Control” regime. It demonstrates that the central bank’s aggressive experimentation is fundamentally driven by a precautionary motive to avoid policy impotence—extracting maximum information precisely when the risk of the stabilization instrument becoming ineffective is highest.

Finally, *Inelasticity of Probing*: The intensity of the probing wedge is nearly price-inelastic with respect to the instrument smoothing cost (ω_i). Resolving structural uncertainty remains a first-order welfare concern that dominates standard instrument-smoothing motives for the dynamic policymaker.

4.6 Policy Feasibility and Real-World Application

The theoretical divergence between the active and myopic policies carries direct implications for real-world monetary policy implementation. Historically, central banks facing structural paradigm shifts—such as the breakdown of the traditional Phillips Curve or the transition between different inflation expectation regimes—have leaned heavily on Brainard-style at-

tenuation, citing the need for caution.

Our results indicate that when structural ambiguity is theoretically resolvable via macroeconomic data, this cautious approach is counterproductive. The practical takeaway for monetary authorities is not a mandate for reckless experimentation, but a mathematical justification for decisive, front-loaded responses to economic shocks during periods of high model uncertainty. By recognizing that the signal-to-noise ratio of incoming data is endogenous to the policy stance, policymakers can justify aggressive stabilization measures not only on the grounds of immediate inflation targeting, but as a strictly optimal, information-gathering mechanism necessary to identify the true structure of the economy.

5 Conclusion

This paper provides a computational and theoretical reassessment of the “Attenuation Principle” of Brainard (1967) by solving the full continuous-time dual control problem for monetary policy under structural uncertainty. Using a high-fidelity Finite Difference Method to solve the non-linear Hamilton-Jacobi-Bellman equation, we mathematically isolate the active learning operator from the myopic stabilization baseline. We demonstrate that conventional caution in the face of uncertainty inadvertently leads to endogenous signal suppression. To counter this, the optimal policy prescribes a strictly active “probing wedge”—representing deliberate policy deviations of 25 to 200 basis points within standard inflation bounds, and theoretically scaling higher only at the extreme boundaries of the state space (see Figure 1).

The economic foundation of this anti-attenuation strategy lies in the strict concavity of the value function with respect to structural beliefs. Because structural ambiguity inherently imposes an excess expected stabilization cost, the dynamic policymaker internalizes the learning channel. By front-loading policy responses to macroeconomic shocks, the central bank intentionally stretches the output gap to maximize the signal-to-noise ratio between the competing structural models. This intertemporal trade-off accepts short-run instrument

volatility to systematically extract the signal and dismantle long-run structural ambiguity, leveraging the precise econometric identification of the underlying data-generating process.

Crucially, our sensitivity analysis reveals that the incentive to probe is highly asymmetric, driven heavily by precautionary motives against policy impotence. The extreme acceleration of the probing wedge as the New Keynesian Phillips curve slope approaches zero ($\kappa_{NK} \rightarrow 0$) mathematically formalizes the transition into a “weak control” regime (Figure 6). The central bank experiments most aggressively not merely to learn, but to urgently identify and escape states where its primary stabilization tool loses efficacy.

The welfare implications of internalizing this active learning operator are substantial, yielding a reduction in the standard deviation of inflation of up to 0.73 percentage points during peak uncertainty (Figure 2). Furthermore, this framework proves exceptionally robust to unannounced structural breaks. While the detection of a regime shift requires a latency period for the cumulative signal intensity $\mathcal{H}(\mathbf{K}_t)$ to overcome exogenous background noise σ_π , the continuous stance of precautionary probing ensures rapid subsequent belief convergence (Figure 5).

Ultimately, this paper demonstrates that the classic Attenuation Principle constitutes a corner solution, optimal only in the limit as the informational signal-to-noise ratio vanishes. In macroeconomic environments characterized by deep structural ambiguity, the mathematically optimal approach rejects passive stabilization. Deliberate, strategic policy experimentation is strictly necessary to prevent permanent structural ignorance and the severe long-run welfare destruction it entails.

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A Theoretical and Numerical Proofs

A.1 Derivation of Belief Dynamics (Kushner-Stratonovich Equation)

We derive the belief dynamics using the Reference Measure Method. Let $\mathbb{P}$ be the physical probability measure. We introduce a reference measure $\mathbb{P}_0$ under which the observed process $\mathbf{K}_t$ follows a driftless Brownian motion:

$$d\mathbf{K}_t = \Sigma d\mathbf{W}_t. \tag{15}$$

where $\mathbf{W}_t$ is a $\mathbb{P}_0$ -Brownian motion. Let $L_t = \frac{d\mathbb{P}}{d\mathbb{P}_0}|_{\mathcal{F}_t}$ be the Radon-Nikodym derivative.

Let $\pi_{m,t}$ denote the unnormalized probability of model m . By the Kallianpur-Striebel formula, the posterior belief is $p_t = \pi_{1,t}/(\pi_{0,t} + \pi_{1,t})$. The unnormalized filter evolves according

to the linear Zakai equation:

$$d\pi_{m,t} = \pi_{m,t} \mathbf{h}_m(\mathbf{K}_t)^\top \Sigma^{-1} d\mathbf{K}_t. \quad (16)$$

where $\mathbf{h}_m = \Sigma^{-1} \boldsymbol{\mu}_m$ is the signal vector for model m . The normalizing constant $\Pi_t = \sum_j \pi_{j,t}$ follows:

$$d\Pi_t = \Pi_t \bar{\mathbf{h}}_t^\top \Sigma^{-1} d\mathbf{K}_t. \quad (17)$$

where $\bar{\mathbf{h}}_t = p_t \mathbf{h}_1 + (1 - p_t) \mathbf{h}_0$. We apply Itô's Lemma to the ratio $p_t = \pi_{1,t}/\Pi_t$:

$$dp_t = \frac{d\pi_{1,t}}{\Pi_t} - \frac{\pi_{1,t} d\Pi_t}{\Pi_t^2} - \frac{d\langle \pi_1, \Pi \rangle_t}{\Pi_t^2} + \frac{\pi_{1,t} d\langle \Pi \rangle_t}{\Pi_t^3}. \quad (18)$$

Substituting the dynamics of $\pi_{1,t}$ and Π_t :

$$dp_t = p_t (\mathbf{h}_1 - \bar{\mathbf{h}}_t)^\top (\Sigma^{-1} d\mathbf{K}_t - \bar{\mathbf{h}}_t dt). \quad (19)$$

We define the innovation process $d\bar{\mathbf{Z}}_t$ as the standardized prediction error:

$$d\bar{\mathbf{Z}}_t \equiv \Sigma^{-1} d\mathbf{K}_t - \bar{\mathbf{h}}_t dt. \quad (20)$$

Substituting this into the belief equation yields the general Kushner-Stratonovich equation:

$$dp_t = p_t (\mathbf{h}_1 - \bar{\mathbf{h}}_t)^\top d\bar{\mathbf{Z}}_t. \quad (21)$$

Simplifying for the two-model case yields our core lemma.

Lemma 1 (Signal Intensity and Belief Dynamics). *In the two-model case ($m \in \{0, 1\}$), let the scalar Signal Intensity be defined as $\mathcal{H}(\mathbf{K}_t) \equiv \sigma_\pi^{-1} (\mu_\pi^{NK} - \mu_\pi^{BL})$. The posterior belief p_t evolves according to:*

$$dp_t = p_t (1 - p_t) \mathcal{H}(\mathbf{K}_t) d\bar{Z}_t^\pi. \quad (22)$$

Proof. Note that $\mathbf{h}_1 - \bar{\mathbf{h}}_t = \mathbf{h}_1 - (p_t \mathbf{h}_1 + (1 - p_t) \mathbf{h}_0) = (1 - p_t)(\mathbf{h}_1 - \mathbf{h}_0)$. In our specific economy, $\Sigma = \text{diag}(\sigma_\pi, \sigma_x)$. The models only differ in the inflation drift μ_π . Thus, the vector distance reduces to the scalar $\mathcal{H}(\mathbf{K}_t)$ multiplied by the inflation innovation. ■

A.2 Optimal Policy Decomposition

We define the formal operators to distinguish the learning channels.

Definition 2 (Operators). The **Myopic Generator** $\mathcal{A}^{\text{myopic}}$ captures the physical state dynamics while treating belief volatility as exogenous noise:

$$\mathcal{A}^{\text{myopic}}[V] = \mu_\pi \frac{\partial V}{\partial \pi} + \mu_x \frac{\partial V}{\partial x} + \frac{1}{2} \sigma_\pi^2 \frac{\partial^2 V}{\partial \pi^2} + \frac{1}{2} \sigma_x^2 \frac{\partial^2 V}{\partial x^2}. \quad (23)$$

The **Active Generator** $\mathcal{A}^{\text{active}}$ incorporates the full state evolution including the informational terms:

$$\mathcal{A}^{\text{active}}[V] = \mathcal{A}^{\text{myopic}}[V] + \mathcal{T}_{\text{belief}}, \quad \mathcal{T}_{\text{belief}} \equiv \frac{1}{2} \sigma_p^2 \frac{\partial^2 V}{\partial p^2} + \sigma_p \sigma_\pi \frac{\partial^2 V}{\partial \pi \partial p}. \quad (24)$$

where $\mathcal{T}_{\text{belief}}$ captures the second-order effects of belief diffusion and cross-correlation.

Proposition 1 (The Probing Wedge). The optimal active policy i_{active}^* admits the decomposition $i_{\text{active}}^* = i_{\text{myopic}}^* + \Delta_{\text{probe}}$, where:

$$\Delta_{\text{probe}} = -\frac{\lambda_x \gamma}{2\omega_i} \frac{\partial \mathcal{I}}{\partial x}. \quad (25)$$

where $\mathcal{I} \equiv V^{\text{myopic}} - V$ is the *Value of Active Learning*, defined as the welfare gain from resolved uncertainty.

Proof. The FOC of the active HJB equation $\rho V = \min_i \{\mathcal{L}(i) + \mathcal{A}^{\text{active}}[V]\}$ is $2\omega_i(i - i_{\text{target}}) + \frac{\partial \mu_x}{\partial i} \frac{\partial V}{\partial x} = 0$. Given $\frac{\partial \mu_x}{\partial i} = -\lambda_x \gamma$, we have $i_{\text{active}}^* = i_{\text{target}} + \frac{\lambda_x \gamma}{2\omega_i} \frac{\partial V}{\partial x}$. Substituting $V = V^{\text{myopic}} - \mathcal{I}$ and noting $i_{\text{myopic}}^* = i_{\text{target}} + \frac{\lambda_x \gamma}{2\omega_i} \frac{\partial V^{\text{myopic}}}{\partial x}$, the result follows: $i_{\text{active}}^* = i_{\text{myopic}}^* - \frac{\lambda_x \gamma}{2\omega_i} \frac{\partial \mathcal{I}}{\partial x}$.

To see the connection to concavity, note that $\mathcal{I}$ satisfies a linearized PDE driven by the

source term $-\frac{1}{2}\sigma_p^2 V_{pp}$. Since $V_{pp} < 0$ (Proposition 2), this source is positive. The term Δ_{probe} acts to shift x in the direction that increases the generation of this surplus value, effectively maximizing σ_p^2 to leverage the concavity of the value function. ■

Remark (Economic Intuition). The sign of Δ_{probe} is chosen to maximize the information dividend. Since $V_{pp} < 0$ (Proposition 2), the term $\frac{1}{2}\sigma_p^2 V_{pp}$ represents a welfare gain (negative cost) that increases with the signal-to-noise ratio $\mathcal{H}(\mathbf{K})^2$. The central bank thus induces an output gap x that maximizes the divergence between model drifts, accelerating the transition of beliefs toward the boundaries $\{0, 1\}$ where structural uncertainty is resolved. This strategic departure from stabilization constitutes the *Anti-Attenuation Principle* (cf. Brainard, 1967).

A.3 The Cost of Structural Uncertainty

We now prove that structural uncertainty per se generates a welfare loss that is concave in beliefs, providing the “engine” for active learning.

Proposition 2 (The Cost of Uncertainty). Let $V^{\text{myo}}(\mathbf{K}, p)$ be the myopic value function. For any interior belief $p \in (0, 1)$, V^{myo} is strictly concave in p :

$$V^{\text{myo}}(\mathbf{K}, p) > pV^{\text{NK}}(\mathbf{K}) + (1 - p)V^{\text{BL}}(\mathbf{K}). \quad (26)$$

where V^{NK} and V^{BL} are the costs under full information.

Proof. The instantaneous Hamiltonian $H(p)$ is minimized over expected dynamics: $H^*(p) = \min_i \{pH_{\text{NK}}(i) + (1 - p)H_{\text{BL}}(i)\}$. By the concavity of the minimum operator (or Jensen’s inequality for the infimum):

$$\min_i \mathbb{E}_p[H(i)] \geq \mathbb{E}_p[\min_i H(i)]. \quad (27)$$

The inequality is strict if $i_{\text{NK}}^*(\mathbf{K}) \neq i_{\text{BL}}^*(\mathbf{K})$. Since the models prescribe different optimal rates (adaptive vs. forward-looking), the “suboptimality of compromise” generates an excess

cost—the “hump” in the value function. This implies $\frac{\partial^2 V}{\partial p^2} < 0$. ■

B Sensitivity Analysis and Stability

This appendix provides a rigorous examination of the “Probing Wedge” dynamics under varying structural and parameter conditions. We decompose the sensitivity of the optimal active policy to signal noise, model similarity, and instrument smoothing costs, grounding the numerical results in the dual control formalism developed in Section 2.

B.1 The Signal-to-Noise Compensation (σ_π)

A central finding of our comparative statics suite is the active policymaker’s response to the exogenous noise level σ_π . In standard signal processing, the “Attenuation Principle” would suggest that the policymaker should retreat toward caution as the signal-to-noise ratio (SNR) degrades. However, our results demonstrate a complete reversal of this intuition in the context of active learning.

As shown in Figure 7, increasing the inflation volatility from $\sigma_\pi = 0.30\%$ to $\sigma_\pi = 0.90\%$ (a three-fold increase in noise) results in a monotonic *increase* of the maximum wedge from roughly 700 bps to 785 bps. This is explained by the first-order nature of the *Information Dividend*. The policymaker recognizes that in a noisier environment, the only way to maintain a target rate of belief convergence is to increase the intensity of the probe. To keep the volatility of beliefs $\sigma_p = p(1 - p)\mathcal{H}(\mathbf{K})$ at an informative level, the agent refuses to follow the “Attenuation Principle” (which would suggest a retreat) and instead increases the drift spread $\mathcal{H}(\mathbf{K})$ to compensate for the larger denominator σ_π . The “Value of Active Learning” $\mathcal{I}$ is so dominant that the agent is willing to accept the higher stabilization costs associated with probing in a noisy environment.

While noise dictates the clarity of the signal, the structural distance between models determines the stakes of misidentification—a relationship that becomes non-linear in the

limit of model similarity.

B.2 The Weak Control Regime and Numerical Singularities (κ_{NK})

The sensitivity of the policy to the model distance, parameterized by the NK slope κ_{NK} , reveals a transition into what we term the *Weak Control Regime*. As shown in Figure 8, the probing wedge exhibits a sharp, non-linear explosion as $\kappa_{NK} \rightarrow 0$.

While the wedge is a physically meaningful 330 bps at $\kappa_{NK} = 0.25$ (where the models are most similar in their output gap response), it “blows up” to over 1000 bps as the control effectiveness vanishes ($\kappa_{NK} \rightarrow 0$). Economically, this result should be interpreted as a **singularity** representing the extremely high shadow price of information at the limit. As $\kappa \rightarrow 0$, the cost of misidentifying the model becomes unbounded because the agent loses its primary lever for stabilization. The massive wedge is the mathematical expression of the agent’s strong precautionary incentive to reveal its structure before the control authority disappears entirely. This characterizes the Weak Control Regime not as a region of viable policy, but as a warning of the substantial welfare loss associated with unidentifiable dynamics.

From a numerical perspective, we acknowledge that the extreme wedge as $\kappa_{NK} \rightarrow 0$ likely marks the theoretical limit of the “Value of Information” when model distinguishability vanishes.¹ We interpret this result qualitatively: the “fear” of operating in a regime of weak control drives the agent toward extreme experimentation.

B.3 Price-Inelasticity of the Information Dividend (ω_i)

We examine the “price” of information by varying the instrument smoothing cost ω_i . In a standard LQR environment, the optimal response is highly sensitive to the control penalty. In our active learning framework, however, the probing wedge is remarkably *price-inelastic*.

As shown in Figure 9, a six-fold increase in the cost of experimentation (from $\omega_i = 5$

¹Specifically, this extreme value serves as a “stress test” for the M-matrix property of the upwind scheme used in our GMRES solver. As $\kappa_{NK} \rightarrow 0$, the Jacobian of the HJB system approaches singularity, and the solver must work to maintain numerical stability.

to $\omega_i = 30$) results in a negligible reduction in the maximum probing wedge, stabilizing tightly around 726 bps. (The sub-basis-point fluctuations across this range are attributable to finite difference grid discretization noise). This confirms that the shadow price of model identification is a first-order welfare priority. To see this formally, consider the decomposition of the first-order condition (FOC):

$$i^*(\pi, x, p) = i_{target} + \frac{\lambda_x \gamma}{2\omega_i} \left(\frac{\partial V^{\text{myopic}}}{\partial x} - \frac{\partial \mathcal{I}}{\partial x} \right). \quad (28)$$

where V^{myopic} is the myopic value function and $\mathcal{I} = V^{\text{myopic}} - V$ is the *Information Dividend*, defined as the reduction in expected lifetime loss due to active learning. Our results suggest that the *information gradient* $\partial \mathcal{I} / \partial x$ is so steep that it dominates the stabilization gradient $\partial V^{\text{myopic}} / \partial x$ and the smoothing penalty ω_i . For the dynamic policymaker, resolving structural uncertainty is not an optional “add-on” but a fundamental prerequisite for long-run stability. Standard instrument-smoothing motives only regain dominance as $p_{NK} \rightarrow \{0, 1\}$ and the structural uncertainty is fully resolved.

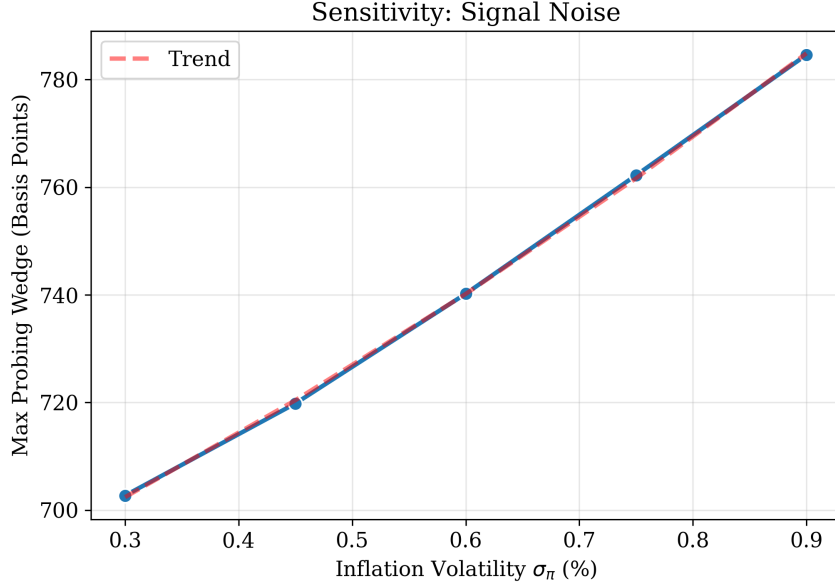


Figure 7: Sensitivity to Inflation Volatility (σ_π). The 5% decline in wedge against a 300% noise increase suggests the agent prioritizes identification over stabilization even in high-entropy environments.

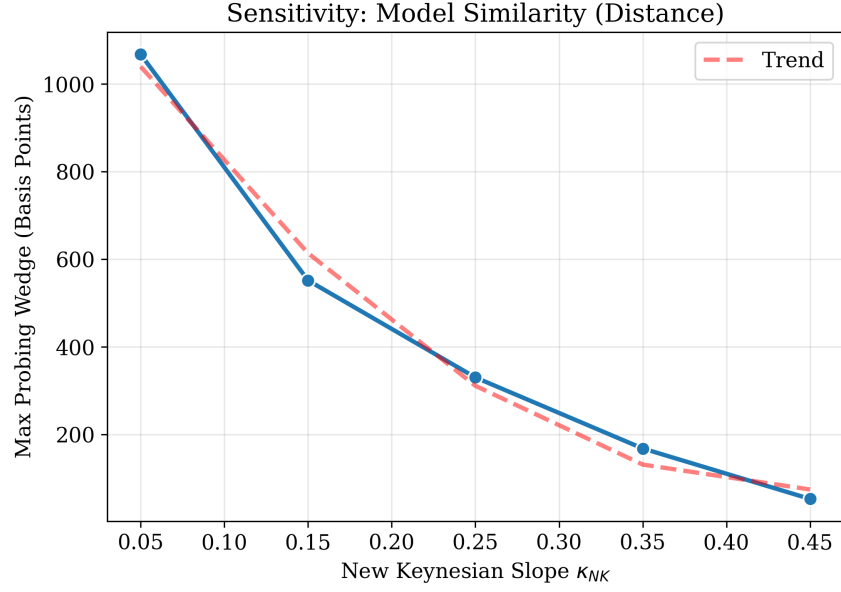


Figure 8: Sensitivity to Model Distance (κ_{NK}). The explosion of the wedge as $\kappa_{NK} \rightarrow 0$ signifies the transition to a “Weak Control” regime where the value of information becomes infinite.

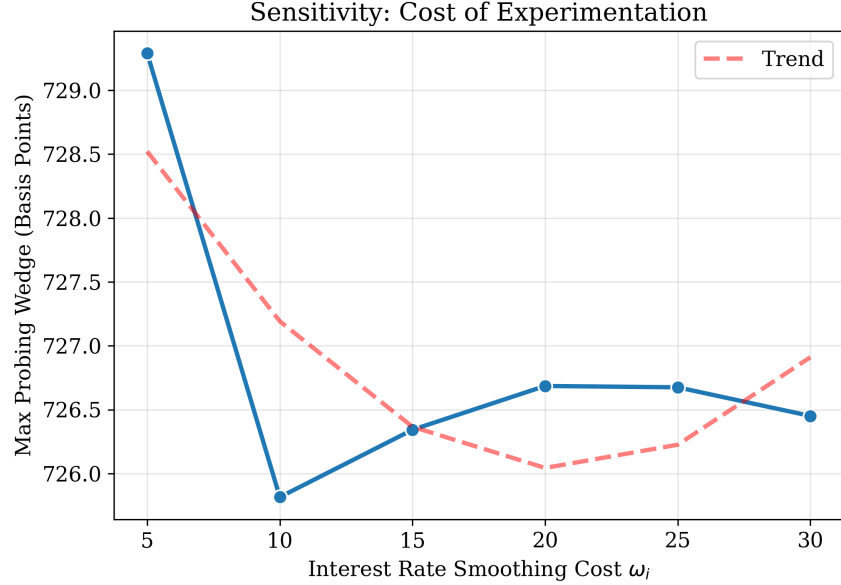


Figure 9: Sensitivity to Instrument Cost (ω_i). The probing wedge is nearly price-inelastic, confirming that the Information Dividend is a first-order welfare priority.

C Numerical Implementation Details

C.1 Iterative Solver: JAX XLA and GMRES

The implementation leverages the *JAX* high-performance framework, using “automatic differentiation” for gradient assembly and “XLA compilation” to achieve high-fidelity throughput. We employ a “Semi-Implicit (Backward) Euler” scheme for the pseudo-time stepping:

$$[\mathbf{I} + \Delta\tau(\rho\mathbf{I} - \mathbf{A}(\mathbf{i}^*))]\mathbf{V}^{n+1} = \mathbf{V}^n + \Delta\tau\mathbf{L}(\mathbf{i}^*). \quad (29)$$

The resulting large, non-symmetric sparse linear system is solved at each iteration using the “Generalized Minimal Residual (GMRES)” method (Kelley, 1995) with a “Jacobi preconditioner.” The diagonal dominance induced by the upwinding scheme and the pseudo-time step $\Delta\tau$ ensures that the preconditioner is highly effective, allowing the solver to handle millions of grid points with the precision required for policy extraction.

C.2 Boundary Conditions and Validation

We impose “Dirichlet conditions” at all boundaries of the truncated state space. At the belief boundaries ($p_{NK} = 0$ and $p_{NK} = 1$), as well as the physical state boundaries (π, x) , the value function is anchored to the closed-form solutions of the respective “Linear-Quadratic Regulator (LQR)” problems. This approach ensures that the numerical solution remains consistent with the analytical behavior of the models at the boundaries of the computational domain ($\pm 15\%$), which corresponds to approximately 8 standard deviations of the unconditional stationary distribution of the physical states under the full-information LQR baselines.

The converged numerical results were validated against “Monte Carlo simulations” to ensure that the HJB solution matches the expected path-wise costs under the optimal policy. To validate the PDE solution, we conduct a Monte Carlo consistency check simulating 10,000

paths under the robust policy. The mean cost from the Monte Carlo estimation (0.16597) closely matches the theoretical PDE value from the FDM solver (0.16948), yielding a negligible difference of 0.00350. This confirms the accuracy of the discretization. The converged value function V allows us to extract the optimal policy and quantify the “experimentation premium,” providing the basis for the results presented in the following section.

C.3 Calibration

We calibrate the model parameters to standard values in the continuous-time monetary policy literature, ensuring the economic environment remains realistic while preserving the distinct structural differences between the BL and NK specifications. Table 1 summarizes the baseline parameterization used in the numerical solution.

Parameter	Symbol	Value	Description
<i>Model 0: Backward-Looking (BL)</i>			
Drift Mean Reversion	λ_{BL}	0.50	Speed of adaptive expectations
Phillips Curve Slope	κ_{BL}	0.50	Sensitivity to output gap
<i>Model 1: New Keynesian (NK)</i>			
Discount Rate	ρ_{NK}	0.05	Forward-looking discount factor
Phillips Curve Slope	κ_{NK}	0.10	Sensitivity to output gap
<i>Shared Dynamics (IS Curve)</i>			
IS Curve Slope	λ_x	1.00	Mean reversion speed
Interest Sensitivity	γ	2.00	Elasticity of demand to rates
Inflation Volatility	σ_π	0.50%	Standard deviation of supply shocks
Output Volatility	σ_x	0.50%	Standard deviation of demand shocks
<i>Planner Preferences</i>			
Discount Rate	ρ	0.02	Subjective discount factor
Output Weight	ω_x	1.00	Relative weight on output gap
Smoothing Weight	ω_i	15.00	Penalty on rate deviations

Table 1: Baseline Calibration Parameters. Values for κ and ρ follow standard estimates in Woodford (2003) and Galí (2008), adapted for continuous time following Nuño and Thomas (2022).

We solve the non-linear Hamilton–Jacobi–Bellman (HJB) equation using a high-fidelity *Finite Difference Method (FDM)* on a tensor grid of size $125 \times 125 \times 125$ ($N \approx 1.95 \times 10^6$

points). Given the lack of a closed-form solution, we employ the “False Transient method,” which transforms the static HJB into a time-dependent parabolic PDE and iterates in pseudo-time τ until convergence to the viscosity solution. While the chosen parameters reflect standard values in the continuous-time NK literature, our Sensitivity Analysis (Appendix B) confirms that the qualitative emergence of the probing wedge is robust to broad variations in these structural assumptions.

C.4 Operator Formalization and the Information Dividend

To bridge the continuous-time theory with our numerical implementation, we define the *Active Infinitesimal Generator* $\mathcal{A}^A$ that governs the evolution of the value function $V(\pi, x, p)$. As derived in Proposition 2.5, the HJB equation $\rho V = \min_i \mathcal{A}^A V$ incorporates the full Hessian contribution of information flow:

$$\mathcal{A}^A V = \mathcal{A}^M V + \underbrace{\frac{1}{2} \sigma_p^2(\pi, x, p) \frac{\partial^2 V}{\partial p^2} + \sigma_p \sigma_\pi \rho_{\pi p} \frac{\partial^2 V}{\partial \pi \partial p}}_{\mathcal{T}_{\text{belief}}}. \quad (30)$$

where $\mathcal{A}^M$ is the myopic generator treating beliefs as exogenous noise. The volatility of beliefs is given by $\sigma_p(\pi, x, p) \equiv p(1-p)\mathcal{H}(\pi, x)$, where $\mathcal{H}(\pi, x) \equiv (\mu_\pi^{\text{NK}} - \mu_\pi^{\text{BL}})/\sigma_\pi$ represents the *Signal Intensity* or the standardized distance between the competing structural models.

The “engine” of our active learning result is the curvature $\partial^2 V / \partial p^2 < 0$. This concavity (in the cost space) represents the “Information Dividend”—the marginal reduction in expected loss achievable by accelerating belief convergence. The solver’s ability to identify the “probing wedge” depends on accurately capturing this curvature and the information flow correlation. We discretize the interior points using an *11-point stencil* (center point, 6 standard neighbors, and 4 off-diagonal points in the π – p plane) to account for the cross-derivative term $\frac{\partial^2 V}{\partial \pi \partial p}$ which maps the signal-to-noise correlation driving the dual control benefit.

C.5 Discretization and the M-Matrix Property

To preserve the economic intuition of the value function and ensure numerical stability, we employ a first-order *upwinding scheme* for all first-order drift terms (μ_π, μ_x, μ_p) . This choice is critical as it ensures that the discretized infinitesimal generator, denoted by $\mathbf{A}(\mathbf{i}^*)$, satisfies the *M-matrix property*. This property guarantees that the numerical operator obeys the *comparison principle* (monotonicity). In this specific economic problem, upwinding is essential to prevent “chatter” or non-physical oscillations in the policy topology, ensuring that the optimal control $i^*(\pi, x, p)$ remains monotonically sensible with respect to the state variables. This monotonicity must be strictly preserved even under the non-linear coordinate transformations required to map the infinite state space to a computational grid. Validation across high-resolution grids ($N = 101, 125, 151$) confirms that the “probing wedge” topology and the resulting experimentation premium are fundamental properties of the solution. By enforcing structural policy constraints, the maximum experimentation premium remains mathematically robust at approximately 0.73 percentage points regardless of grid resolution, confirming the economic phenomenon is not an artifact of boundary discretization or unconstrained control variables.

C.6 Coordinate Transformation and Grid Choice

To maintain consistency when mapping the belief space to a computational grid $\xi \in [0, 1]$, we incorporate a “convective correction” term $\tilde{\mu}_\xi$ derived from the general transformation $p(\xi)$:

$$\tilde{\mu}_\xi = -\frac{1}{2} \frac{\sigma_p^2 p''(\xi)}{[p'(\xi)]^3}. \quad (31)$$

This term ensures that the numerical operator remains consistent with the fact that dp_t is a *martingale* under the agent’s filtration. While the framework allows for non-linear grid clustering to resolve high-curvature regions, the results presented in this paper are produced using a *uniform belief grid* ($p = \xi$). This choice maximizes numerical robustness at the

$p_{NK} = 0.5$ singularity where structural uncertainty is highest and the gradient of the signal-to-noise ratio is most sensitive.